

# LAB 1 - REQUIREMENTS ENGINEERING

Student / SRN  
Project  
Course / Lab

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Hospital Bed & ICU Allocation Dashboard  
Software Engineering - Lab 1

## Requirements Table

| Req ID  | Type          | Description                                                                                             | Priority | Acceptance Criteria                                                                                                                                                                      | Rationale                                                                                     | Comments / Review               |
|---------|---------------|---------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------|
| FR-001  | Functional    | The system shall allow the triage nurse to view the current availability of hospital beds and ICU beds. | High     | When the triage nurse opens the availability dashboard, the system shall display the current count and status of available, occupied, reserved and unavailable beds for each ward/ICU.   | Bed availability is essential for making an appropriate allocation decision.                  | Add feedback after peer review. |
| FR-002  | Functional    | The system shall allow the triage nurse to allocate an available bed or ICU bed to a patient.           | High     | When an available bed is selected and the allocation is confirmed, the system shall associate the bed with the patient and change the bed status from Available to Occupied or Reserved. | Bed allocation is the primary function of the system.                                         | Add feedback after peer review. |
| FR-003  | Functional    | The system shall allow authorized hospital staff to update the status of a hospital bed or ICU bed.     | High     | When authorized staff update a bed status, the system shall save the new status and display the updated status on the dashboard within 5 seconds.                                        | Accurate bed status is necessary to prevent incorrect allocation.                             | Add feedback after peer review. |
| FR-004  | Functional    | The system shall notify the triage nurse when no suitable bed or ICU bed is available for a patient.    | High     | When the requested bed category has zero available beds, the system shall display an unavailable notification and show any currently available alternative bed categories.               | Nurses need immediate visibility of alternatives when the requested resource is unavailable.  | Add feedback after peer review. |
| FR-005  | Functional    | The system shall allow the hospital administrator to view bed occupancy and allocation information.     | Medium   | When the hospital administrator opens the dashboard, the system shall display the number of total, available, occupied, reserved and unavailable beds and ICU beds.                      | Administrators require an overall view of hospital capacity for operational decisions.        | Add feedback after peer review. |
| NFR-001 | Nonfunctional | The system shall display the latest bed availability information within 3 seconds of a user request.    | High     | In at least 95 out of 100 availability requests under normal operating load, the dashboard shall display the requested bed availability information within 3 seconds.                    | Fast availability information is important during patient admission and emergency situations. | Add feedback after peer review. |
| NFR-002 | Nonfunctional | The system shall ensure that only authorized users can perform bed allocation and bed-status updates.   | High     | When an unauthorized user attempts to allocate a bed or modify its status, the system shall reject the operation and display an access-denied message.                                   | Unauthorized changes could result in incorrect or unsafe bed allocation information.          | Add feedback after peer review. |

**Peer critique requirement:** During the lab, have a classmate review clarity, testability and relevance. Replace the placeholder comments above with the actual feedback and revise requirements where needed before final submission.